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Fourier Ptychographic Microscopy Reconstruction Method Based on Residual Transfer Networks

Xiaoli Wang^{1,2a}, Yan Piao^{1*b}, Jie Li^{2c}, Jinyang Yu^{2d}

¹Changchun University of Science and Technology Electronics Information Engineering College, Changchun, China

²Changchun University Electronics Information Engineering College, Changchun, China

^awangxl@ccu.edu.cn ^{*b}1006285507@qq.com ^cLij69@ccu.edu.cn

^d1564152488@qq.com

Abstract. Fourier ptychographic microscopy reconstruction mostly adopts the traditional alternating iterative phase recovery method and optimization method, which has high computational complexity, high redundancy of image acquisition data, low reconstruction quality and high time consumption. In this paper, the model of residual transfer networks based on Resnet152 is proposed for Fourier ptychographic microscopy reconstruction, the learning process of deep convolution neural network is introduced, and the image reconstruction method based on deep learning realizes the end-to-end reconstruction of low-resolution images to high-resolution images. Through comparative experiments and analysis, the residual network can overcome the gradient explosion, make the feature information more complete and efficient, and the incremental up-sampling reconstruction network has higher image quality, lower computational complexity and shorter running time.

1. Introduction

Fourier ptychographic microscopy (FPM) reconstruction is to reconstruct the high-resolution (HR) phase and intensity images of the target sample by using multiple low-resolution (LR) intensity images, which is called phase iterative recovery process. The lost phase information is recovered by using the LR intensity image collected by the FPM system. That is, a series of LR images are used to find the HR complex amplitude that fits the constraint. As shown in Figure 1[1], the reconstruction method for phase iterative recovery.

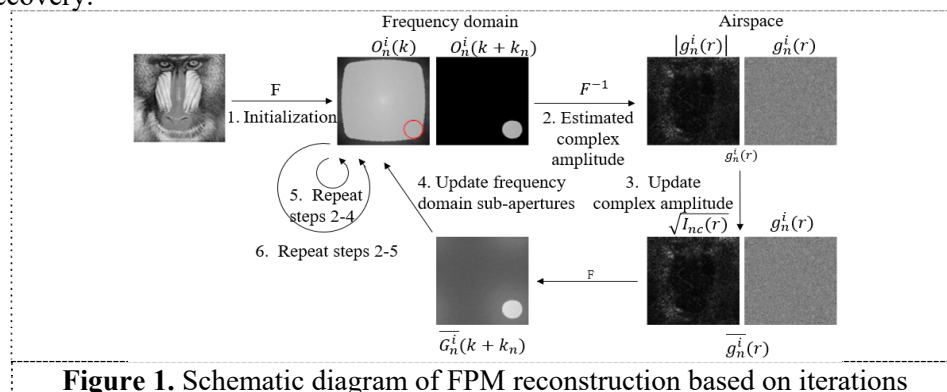


Figure 1. Schematic diagram of FPM reconstruction based on iterations

Recently, researchers have used deep convolutional neural networks as a method for image reconstruction within the field of FPM [2-3]. Jiang et al. [2] proposed a FPM method based on neural network modeling, which is still an iterative algorithm in nature, and the reconstruction speed is still not ideal. Sun et al. [3] combined the residual network structure and deep convolutional network to build an FPM reconstruction network and used a composite loss function to predict the error between the reconstructed result and the true value. Compared with traditional reconstruction algorithms, the network model can quickly reconstruct HR complex amplitude information, and improve the reconstruction efficiency of the FPM system.

In order to better improve the reconstructed image quality and reconstruction speed of the FPM reconstruction method and obtain deep-level features, various high-quality network model frameworks for transfer learning have also emerged [4-5]. It is proved that the convolutional neural network based on transfer learning is better. Therefore, this paper proposes a model based on the transfer residual convolutional network ResNet152 for FPM reconstruction. Compared with the traditional algorithm, the reconstructed image quality and reconstruction speed of the FPM reconstruction method have been greatly improved.

2. Principle

2.1. Transfer the base layer network of ResNet152

The transfer residual convolutional network selects the first 151 layers of ResNet152 [6], freezes the classification layer of the network structure framework, and transfers the network structure of the base network layer of ResNet152, as shown in Figure 2.

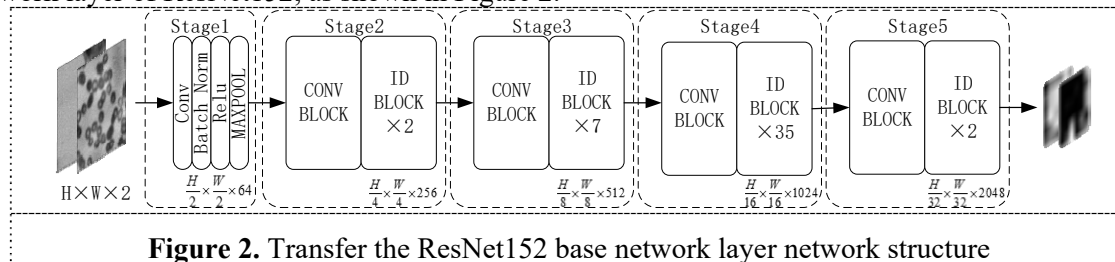


Figure 2. Transfer the ResNet152 base network layer network structure

The network structure of the ResNet152 base network layer mainly consists of five parts: Part 1 Stage1, including convolution (Conv), regularization (Batch Norm), activation function (ReLU), and maximum pooling (MAXPOOLING); the second, third, fourth, and fifth parts, mainly including CONV BLOCK and ID BLOCK residual network modules.

CONV BLOCK residual module, mainly by the calculation process of CONV2D, Batch Norm, ReLU and forward feedback coefficients of CONV2D and Batch Norm, and output through the ReLU activation function. CONV BLOCK does not change the size of the image data.

The ID BLOCK residual module is mainly by the calculation process of CONV2D, Batch Norm, ReLU and forward feedback coefficients of 1, and output through the ReLU activation function. ID BLOCK changes the size of the image data for dimensionality reduction of image features.

H and W represent the length and width of the image data, and ID BLOCK ($\times 2$), ID BLOCK ($\times 7$), and ID BLOCK ($\times 35$) respectively represent the number of residual modules of the ID BLOCK network structure referenced by the transfer of the ResNet152 base layer network structure.

2.2. Incrementally upsampling reconstruction network

Figure 3 shows the structure of the incremental upsampling reconstruction network [7]. Through a multi-layer incremental module (conv-upsample), the size of the feature image can be enlarged, so that the reconstructed input image is closer to the true value of the HR image.

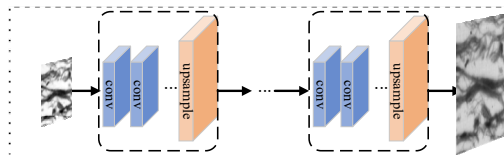


Figure 3. Incremented upsampling schematic diagram

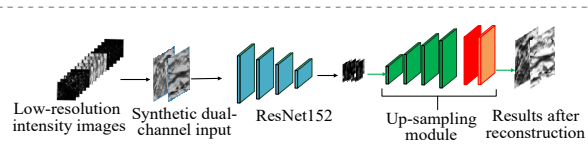


Figure 4. Transfer residual network reconstruction model

2.3. Transfer residual network reconstruction model

Reconstruction of Fourier ptychographic microimaging images based on the residual transfer network model and the network model is shown in Figure 4 with five parts. First, a series of LR intensity images are collected by an FPM system; after the reconstruction process, obtaining dual-channel complex amplitude intensity and phase images by combining acquired data; and then feed the synthesized two-channel data into the transfer ResNet152 network. The purpose of this section is to extract the image features from the input image; then, the extracted features are iteratively upsampled to reconstruct the HR intensity and phase information.

2.4. Transfer residual network structure

The transfer residual network is mainly composed of down-sampling ResNet152 and up-sampling, as shown in Figure 5. Down-sampling ResNet152 performs feature extraction by convolution of input image data (convolution kernel size is 64, convolution size is 7×7), ReLU activation function, and Normalization regularization. Finally, the features are extracted by down-sampling CONV BLOCK and ID BLOCK with a convolution kernel of 256, and the feature channel is 2048; The incremental upsampling part uses the Conv Transpose deconvolution with a convolution kernel of 2048 to complete the feature reconstruction, and output the results of the two-channel reconstruction.

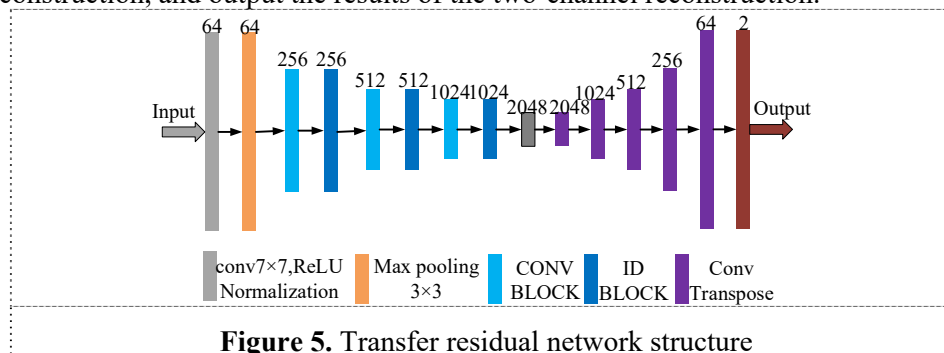


Figure 5. Transfer residual network structure

3. Experiment and results analysis

The experimental test platform is AMD Ryzen9 5900X CPU, the main frequency is 3.70GHz, 64-bit operating system, 64G memory computer, the graphics processor GPU uses the NVIDIA GeForce RTX 3070 operating system for Window10. The programming language is python3.X and open source TensorFlow framework.

3.1. Experimental test data

The image data used in the experimental test is shown in Figure 6. Blood smears and plant tissue sections were selected as test samples. Blood smears containing a single cell showed higher clarity, while plant tissue sections had more texture detail. Figure (a) is the amplitude images, and Figure (b) is the corresponding phase images, and the images' size is all. To ensure experimental contrast, test images were acquired under the same imaging system parameters.

The noise of the FPM system mainly comes from the intensity fluctuation of the LED elements, while the main source of the Gaussian noise is the sensor noise caused by the light source during the image acquisition. Therefore, the collected image can be added to the Gaussian noise to simulate the real noise

and verify the robustness of the reconstruction method. Gaussian noise with a standard deviation of 3×10^{-4} was added to the test image, as shown in Figure 7.

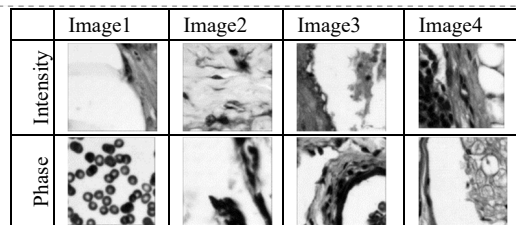


Figure 6. Experimental images

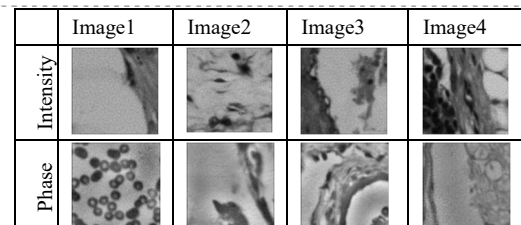


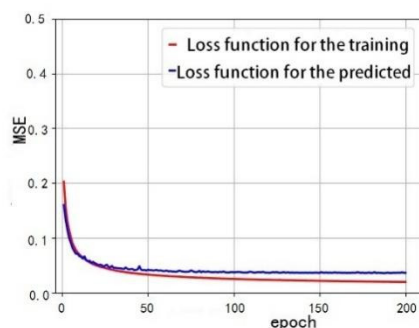
Figure 7. Gaussian noise experimental images

3.2. Stability experiment of the network model

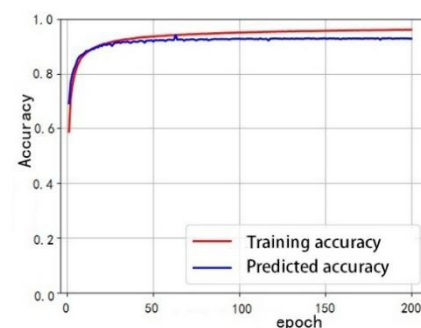
In order to verify the stability of network model training, the loss function and accuracy indicators are used to measure the curve of network model training and predictor index value fitting for different iterations.

The stability curves of training and prediction of the reconstructed network model are shown in Figure 8. Graph (a) is the curve of the training iteration number and loss function, and graph (b) is the curve of the training iteration number and accuracy.

According to the figure, as the number of iterations increases, the training and prediction loss functions of the network also increase. The closer the two curves are, the better the stability of the network model is. The higher the predicted value, the better the reconstruction results. The results show that the proposed network model is stable and feasible.



(a) Epoch and loss function graph



(b) Epoch and accuracy graph

Figure 8. Stability evaluation of the model-fitting curves

3.3. Comparative experiments with traditional algorithms

For the same test image with Gaussian noise, three reconstruction algorithms of GS, AS and Zhang are used to reconstruct the noise image.

Figure 9 shows the reconstruction results of each reconstruction method. It can be seen that the quality of the traditional alternating projection method (GS) and adaptive step size (AS) reconstruction methods is poor, the reconstruction quality of Zhang and the proposed method is higher than that of the traditional method, but the FPM reconstruction method of the transfer residual network in this paper is clearer than that of Zhang's reconstruction algorithm, and there is no noise in the reconstructed image.

Image		Noise Image	GS	AS	Zhang	Ours
Groud1	Intensity					

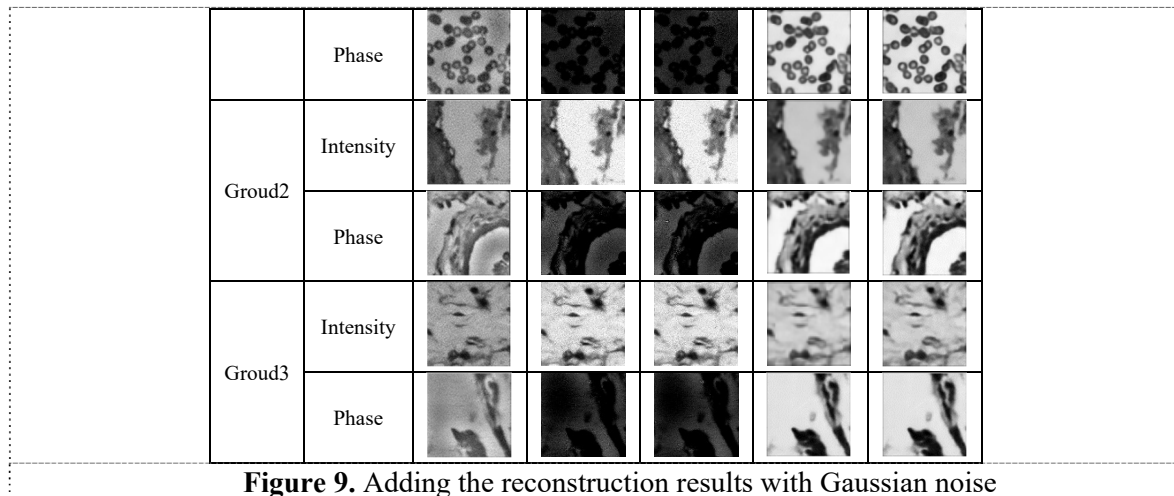


Figure 9. Adding the reconstruction results with Gaussian noise

Peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are used to compare the quality of reconstruction. The data of each reconstruction method are shown in Table 1. In Table 1, when noise is added to the image, the PSNR / SSIM index value of the reconstruction method in this paper is the highest. Compared with the three groups of test images, the PSNR and SSIM values of the reconstructed images are higher than those of AS algorithm, GS algorithm and Zhang algorithm, and the running time is the shortest.

Table 1. Images contain image reconstruction results from different methods with Gaussian noise

Image	Method	Ample image (PSNR/SSIM)	Phase image (PSNR/SSIM)	Time(s)
Groud1 Images	AS	23.5535/0.5469	0.4782/0.1395	1.5430
	GS	23.5619/0.5439	4.7827/0.1391	2.2280
	Zhang	25.1219/0.7225	18.7817/0.8612	201.3
	Ours	26.7164/0.8387	22.5168/0.9186	0.0750
Groud2 Images	AS	23.9409/0.6411	7.2675/0.2098	1.4980
	GS	23.9181/0.6404	7.2831/0.2123	2.1940
	Zhang	24.7164/0.8387	23.6435/0.7139	204.6
	Ours	25.3119/0.8642	26.2516/0.8480	0.0860
Groud3 Images	AS	22.3943/0.5706	4.7791/0.1554	1.6200
	GS	22.4576/0.7749	4.7749/0.1577	2.2350
	Zhang	28.5247/0.8912	25.9134/0.9008	203.4
	Ours	30.7950/0.9235	29.2121/0.9254	0.0840

3.4. Reconstruction experiments of different numbers of images

The reconstructed quality curves under different numbers of images are shown in Figure 10. Red and yellow represent the PSNR and SSIM curves of intensity and phase quality reconstructed by the FPM imaging reconstruction method based on multi-convolution network feature fusion.

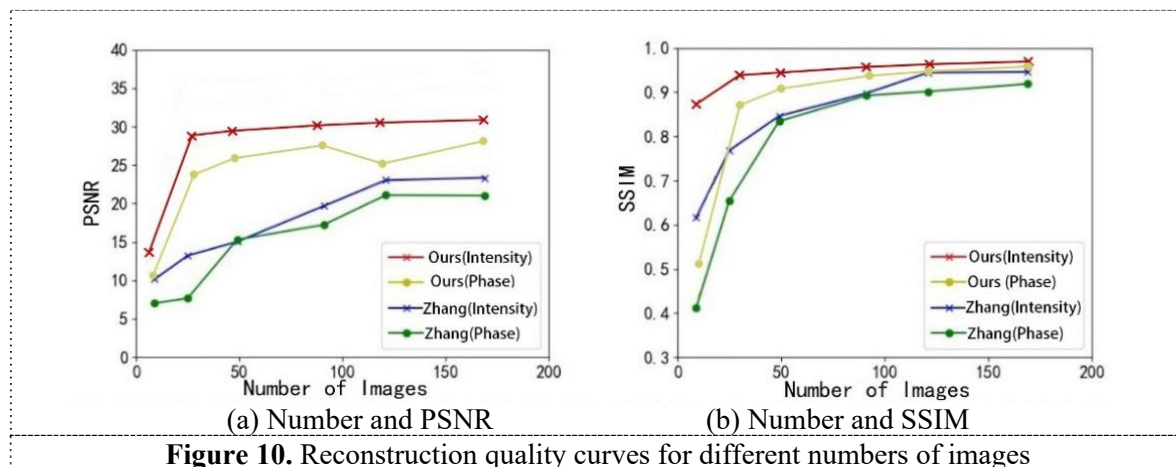


Figure 10. Reconstruction quality curves for different numbers of images

From the reconstructed quality curves, that the intensity and phase PSNR curves of the reconstruction results of the FPM reconstruction method proposed in this chapter are optimal, and the SSIM curve of the reconstructed results in this chapter is better and closer to the true value.

From the reconstruction quality curve, it can be seen that the intensity and phase PSNR curves of the reconstruction method in this chapter are better, and the SSIM curve is better and closer to the true value.

4. Conclusions

The FPM reconstruction method based on the transfer residual network is proposed in this paper. The transfer ResNet152 network is used for feature extraction. Through experiments and comparative analysis, the residual network can overcome the gradient explosion and make the feature information completer and more efficient. The incremental upsampling reconstruction network has good image quality, low computational complexity and short running time.

Acknowledgments

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